CS341 · COMPUTER GRAPHICS

Output Primitives up to DDA

Midterm summary · Chapter 3 & 5

Muharib

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01 Output primitives and OpenGL

Graphics output primitives: functions in a graphics package used to describe the various picture components. Primitives that describe the geometry of objects are called **geometric primitives**.
Basic primitives: points, straight lines, circles, conic sections, curves, surfaces, polygons and filled color areas.

OpenGL is a cross-platform (hardware-independent) 3D graphics library that uses graphics hardware; it is a CG API. It is divided into three libraries:

Library	Files	Header	Prefix
OpenGL (core)	opengl32.lib / .dll	gl.h	gl
Utility (glu)	glu32.lib / .dll	glu.h	glu
Toolkit (glut)	glut32.lib / .dll	glut.h	glut

- Function names start with gl: glBegin, glClear, glVertex.
- Constants: GL_POINTS, GL_LINES, GL_RGB, GL_POLYGON.
- Data types: GLint, GLfloat, GLdouble, GLboolean.
- glut is not part of standard Windows; it is downloaded for free.

02 glVertex and glBegin

```
glVertex2i(4, 5);            2 integer coordinates (z = 0)
```

```
glVertex3f(2.0, 4.1, 6.0);   3 float coordinates
```

```
glVertex3fv(vector);         v = pass an array
```

Vertices are specified between glBegin(mode) and glEnd(), usually counterclockwise for polygons. The current color (glColor3f) applies to every following vertex. The default color is white and the default point size is one pixel.

Type	Modes
Points	GL_POINTS
Lines	GL_LINES, GL_LINE_STRIP, GL_LINE_LOOP
Triangles	GL_TRIANGLES, GL_TRIANGLE_STRIP, GL_TRIANGLE_FAN
Quads	GL_QUADS, GL_QUAD_STRIP
Polygons	GL_POLYGON

With five points p1 … p5:

Mode	Result
GL_LINES	Lines p1–p2 and p3–p4; p5 is ignored
GL_LINE_STRIP	Connected lines p1–p2–p3–p4–p5 (polyline)
GL_LINE_LOOP	Like the strip, plus a closing line p5–p1

03 A complete OpenGL program (slide example)

```
#include <GL/glut.h>
void init (void) {
  glClearColor (1.0, 1.0, 1.0, 0.0);      // window color white (4th =
alpha)
  glMatrixMode (GL_PROJECTION);           // set projection parameters
  gluOrtho2D (0.0, 200.0, 0.0, 150.0);    // 2D orthogonal projection
}
void lineSegment (void) {
  glClear (GL_COLOR_BUFFER_BIT);          // clear the display window
  glColor3f (1.0, 0.0, 0.0);              // red
  glBegin (GL_LINES);
    glVertex2i (180, 15);
    glVertex2i (10, 145);
  glEnd ( );
  glFlush ( );                            // process all routines quickly
}
void main (int argc, char** argv) {
  glutInit (&argc, argv);
  glutInitDisplayMode (GLUT_SINGLE | GLUT_RGB);
  glutInitWindowPosition (50, 100);       // top-left corner of the window
  glutInitWindowSize (400, 300);          // width, height
  glutCreateWindow ("An Example OpenGL Program");
  init ( );
  glutDisplayFunc (lineSegment);
  glutMainLoop ( );                       // display everything and wait
}
```

04 Coordinate systems

System	Meaning
World coordinates	A Cartesian frame where objects are described (2D or 3D). A line is two endpoints; a polygon is its vertices.
Screen coordinates	Integers = pixel positions in the frame buffer. y is the scan-line number and x the column. Hardware counts from the top-left corner.

Low-level functions: setPixel(x, y) stores the current color at that pixel, and getPixel(x, y, color) reads it. In 3D a depth value is stored too; in 2D the depth is 0.

Absolute vs Relative

Absolute	Relative
The values are the actual positions in the coordinate system (relative to the origin).	An offset from the last referenced position (the current position).

```
current position = (3, 8),  relative (2, −1)  →  absolute (5, 7)
```

Repeats in the midterm

What is the difference between absolute and relative coordinates? (472, 461)

gluOrtho2D

```
glMatrixMode (GL_PROJECTION);
```

```
glLoadIdentity ();
```

```
gluOrtho2D (xmin, xmax, ymin, ymax);
```

Sets a 2D frame with `(xmin, ymin)` at the lower-left and `(xmax, ymax)` at the upper-right. Primitives inside the limits are shown and parts outside are clipped. All positions must be absolute.

Coordinate extents: the minimum and maximum x, y, z of an object, also called its bounding box (bounding rectangle in 2D).

05 The line equation

A raster screen cannot draw a perfect line; each point is rounded to the nearest pixel, so every line except horizontal and vertical ones looks stair-stepped (jaggies).

$$y = m \cdot x + b$$

$$m = \frac{yend - y0}{xend - x0}$$

$$b = y0 - m \cdot x0$$

$$\delta y = m \cdot \delta x \qquad \delta x = \frac{\delta y}{m}$$

Slope	Step in
$\|m\| < 1$	x by one unit, compute y
$\|m\| > 1$	y by one unit, compute x
$m = 1$	$\delta x = \delta y$

06 The DDA algorithm

DDA (Digital Differential Analyzer): floating-point increments, then rounded to integers. Rounding of floating-point values makes pixels **drift away** from the true line for long segments, and rounding is **time consuming**.

Case 1: $|m| \leq 1$ x is the variable: $\Delta x = 1 \rightarrow y(k+1) = y(k) + m$

Case 2: $|m| > 1$ y is the variable: $\Delta y = 1 \rightarrow x(k+1) = x(k) + \frac{1}{m}$

Steps

$dx = xEnd - x0$ $dy = yEnd - y0$

$steps = max(|dx|, |dy|)$

$xInc = \frac{dx}{steps}$ $yInc = \frac{dy}{steps}$

plot round(x0), round(y0)

repeat steps times: x += xInc, y += yInc, plot (round(x), round(y))

round(a) = int(a + 0.5)

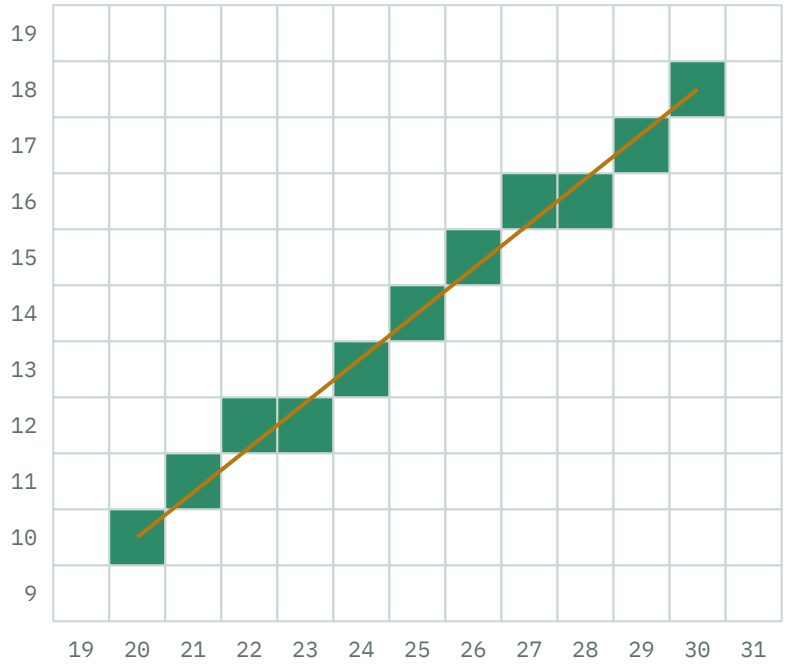
```
void lineDDA (int x0, int y0, int xEnd, int yEnd) {
  int dx = xEnd - x0, dy = yEnd - y0, steps, k;
  float xIncrement, yIncrement, x = x0, y = y0;
  if (fabs (dx) > fabs (dy)) steps = fabs (dx);
  else steps = fabs (dy);
  xIncrement = float (dx) / float (steps);
  yIncrement = float (dy) / float (steps);
  setPixel (round (x), round (y));
  for (k = 0; k < steps; k++) {
    x += xIncrement;  y += yIncrement;
    setPixel (round (x), round (y));
  }
}
```

Slide example: (20,10) to (30,18)

$$dx = 10, \quad dy = 8, \quad m = \frac{8}{10} = 0.8 \quad \rightarrow \quad |m| < 1$$

$$steps = 10, \quad xInc = 1, \quad yInc = 0.8$$

k	x	y	round(x)	round(y)
0	20	10	20	10
1	21	10.8	21	11
2	22	11.6	22	12
3	23	12.4	23	12
4	24	13.2	24	13
5	25	14	25	14
6	26	14.8	26	15
7	27	15.6	27	16
8	28	16.4	28	16
9	29	17.2	29	17
10	30	18	30	18



Midterm 472: (2,2) to (4,10), slope above 1

$$dx = 2, \quad dy = 8, \quad m = \frac{8}{2} = 4 \quad \rightarrow \quad |m| > 1$$

$$steps = 8, \quad xInc = \frac{2}{8} = 0.25, \quad yInc = 1$$

k	x	y	round(x)	round(y)
0	2	2	2	2
1	2.25	3	2	3
2	2.5	4	3	4
3	2.75	5	3	5
4	3	6	3	6
5	3.25	7	3	7
6	3.5	8	4	8
7	3.75	9	4	9
8	4	10	4	10